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rsc.li/catalysisCatalytic upgrading of ethanol to higher alcohols
over nickel-modified Cu–La₂O₃/Al₂O₃ catalysts†Jinfeng He,^a Xiuzhen Li,^b Jianyao Kou,^a Tingjie Tao,^a Xinyue Shen,^a
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A series of nickel and nickel free modified Cu–La₂O₃/Al₂O₃ catalysts were prepared using a simple wetness impregnation method and applied for the catalytic upgrading of ethanol to higher alcohols. 56.7% ethanol conversion, 76.1% total selectivity to higher alcohols and 43.1% total yield thereof were achieved over the optimal 0.65%NiO–Cu–La₂O₃/Al₂O₃ catalyst. The addition of an appropriate amount of NiO effectively promoted ethanol upgrading to higher alcohols by increasing the amount of surface Cu active species and optimizing the acid–base properties of Cu-based catalysts, thus obviously enhancing the dehydrogenation/hydrogenation capacity, as well as the activity of aldol condensation of the catalysts. This study will provide inspiration or direction for the design of more efficient Cu-based catalysts for the catalytic upgrading of ethanol.

Introduction

Bioethanol has attracted extensive attention as an alternative to traditional fossil fuels in recent years, particularly because of concerns over energy security and climate change. However, some limitations are still attached to the usage of ethanol as a fuel additive owing to the low energy density, high water solubility and corrosive behavior of ethanol.¹ In contrast, *n*-butanol as a biofuel has obvious advantages over ethanol and is considered an ideal biofuel or fuel additive.^{2,3} In the last decades, it has been proved that *n*-butanol could be obtained by catalytic upgrading of ethanol *via* the Guerbet pathway (Scheme S1†), which generally proceeds *via* four successive steps: 1) ethanol dehydrogenation, 2) aldol condensation of acetaldehyde, 3) dehydration of 3-hydroxybutyraldehyde, and 4) hydrogenation of crotonaldehyde.⁴ Other higher alcohols such as 2-ethyl-1-butanol, 1-hexanol, 2-ethyl-1-hexanol, and 1-octanol could also be produced simultaneously during ethanol upgrading through a similar mechanism. Different kinds of catalysts, including metal oxides or mixed metal oxides,^{5–8} basic zeolites,⁹ and hydroxyapatite (HAP),^{10–13} have been investigated in the last decades, but their catalytic performance was unsatisfactory even under harsh reaction conditions. Incorporation of transition metals into porous

acid–base materials has been proven to significantly increase the yields of higher alcohols, mainly because the dehydrogenation/hydrogenation steps, which were accomplished previously with acid–base centers, are greatly improved.^{14,15} Among the catalysts containing transition metals, both Cu-based and Ni-based catalysts are promising systems for ethanol upgrading since Cu and Ni are not only well-known active components for ethanol dehydrogenation but also have excellent hydrogenation capability, which can further boost the tandem catalytic process. Our group had demonstrated that Cu–CeO₂/AC catalysts could exhibit an ethanol conversion of 46.2%, a selectivity to *n*-butanol of 41.3% and a total selectivity to C₄–C₈ alcohols of 65% at 523 K and 2 MPa,^{16,17} while ordered mesoporous Cu–La–Al composite metal oxides exhibited an ethanol conversion of 52.2%, a selectivity to *n*-butanol of 51.9% and a total selectivity to higher alcohols of 72.2%.¹⁸ Yang *et al.* firstly investigated the one-step synthesis of *n*-butanol from ethanol over nickel supported gamma alumina catalysts, which exhibited an ethanol conversion of 19.1% and a selectivity to *n*-butanol of 64.3% at 473 K.¹⁹ Ghaziaskar and Xu also investigated the performance of Ni/γ-Al₂O₃ with different nickel contents of 8, 17 and 27 wt% and achieved 35% ethanol conversion and 83% C₄+ alcohol selectivity at 250 °C and 176 bar.²⁰ Unfortunately, due to the strong C–C cleavage ability of metal Ni, there were always CH₄, CO, CO₂ and other cracking products detected in the tail gas, which led to a lower yield of liquid products.²¹ On the other hand, the Cu based catalyst had higher ethanol dehydrogenation activity and zero cleavage capability of the C–C bond under normal reaction conditions but with relatively low hydrogenation activity.²² In

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the current study, the addition of nickel oxide was found to be an effective method to further improve the catalytic performance of Cu-based catalysts for ethanol upgrading to higher alcohols. Furthermore, a nickel promoter at low loadings, mainly in the form of highly dispersed nickel oxide, showed an extremely low capability of cracking the C–C bond of ethanol, acetaldehyde, or other reaction intermediates, thereof avoiding the relatively large loss of liquid products. The action mechanism of a nickel oxide promoter was subsequently investigated by a series of characterization techniques such as N₂-adsorption, XRD, HRTEM, H₂-TPR, XPS, NH₃-TPD and CO₂-TPD. This study will provide inspiration or direction for the design of more efficient Cu-based catalysts for the catalytic upgrading of ethanol.

Experimental

Catalyst preparation

All of chemicals used in this experiment were of analytical grade and used without further purification. A commercial alumina support was purchased from Shandong Aluminum Industry Co., Ltd.

The xNiO–Cu–La₂O₃/Al₂O₃ catalysts, where *x* (0.32–2.55%) represented the mass fraction of NiO in the catalysts, were prepared by a simple wetness impregnation method. The Cu loadings of all the catalysts were fixed at 6 wt%, while the Cu/La molar ratios were 3:1.5. In a typical procedure, stoichiometric nitrate precursors were dissolved in 10 mL ethanol absolute, and a 2 g Al₂O₃ support was added into the above solution. After impregnating at room temperature overnight, the resultant composite was dried in an electric thermostatic drying oven at 383 K for 4 h and subsequently calcined at 723 K for 3 h under an air atmosphere using a muffle furnace. Cu–La₂O₃/Al₂O₃, NiO/Al₂O₃ and NiO–La₂O₃/Al₂O₃ catalysts were prepared using a similar method as reference.

Catalysts characterization

The BET surface areas and the parameters of the pore structure of catalysts were determined by N₂ adsorption using a Quantachrome ASiQC0 FV100-2 apparatus with liquid-N₂ at the temperature of 77 K. The samples were outgassed at 473 K for 4 h prior to analysis.

X-ray diffraction (XRD) data were collected on an X'Pert PRO X-ray diffractometer between $2\theta = 10^\circ$ and 80° at 2° min^{-1} employing a Cu–K α radiation source ($\lambda = 1.5406 \text{ \AA}$). The average crystal diameters of the nanoparticles were calculated using the Scherrer formula.

High resolution transmission electron microscopy (HRTEM) was performed using a Talos S-FEG operated at 300 kV from Philips-FEI Company. Before microscopy examination, the samples were dispersed onto a copper grid coated with a thin holey carbon film.

H₂ temperature-programmed reduction (H₂-TPR) and H₂ temperature-programmed desorption (H₂-TPD) were performed on a Micromeritics Autochem II 2920 chemisorber

equipped with a thermal conductivity detector (TCD) and an OmniStar GSD320 mass spectrometer (MS). For H₂-TPR, the sample (100 mg) was pretreated in a quartz U-tube reactor with a He gas stream (30 mL min^{-1}) at 473 K for 1 h. After the temperature decreased to 323 K, a 10% H₂–Ar flow (50 mL min^{-1}) was introduced, and the sample was heated from room temperature to 1273 K (10 K min^{-1}), while desorbed species were monitored using the TCD and MS in real time. For H₂-TPD, the catalyst was first pre-reduced at 523 K (10 K min^{-1}) in a flow of 10% H₂–Ar (50 mL min^{-1}) for 1 h, purged with Ar (40 mL min^{-1}) at the same temperature for 1 h, and then cooled down to 323 K in an Ar atmosphere. The H₂ uptake of the reduced catalyst was achieved by introducing 10% H₂–Ar till H₂ adsorption saturation. After the H₂ adsorption, the catalyst was purged again with Ar for 30 min at 323 K; then the temperature was linearly increased from 323 K to 1073 K at a rate of 10 K min^{-1} , while the H₂-TPD profiles were recorded with the TCD and MS in real time.

X-ray photoelectron spectroscopy (XPS) spectra were obtained with an ESCALab220i-XL electron spectrometer from VG Scientific using 300 W AlK α radiation and deconvoluted using XPS PEAK41 software. The catalyst after the reaction was transferred into the XPS chamber under Ar protection. The binding energies (BE) were referenced to the graphitic C 1s peak at 284.6 eV, providing accuracy within $\pm 0.2 \text{ eV}$.

CO₂ temperature-programmed desorption (CO₂-TPD) and NH₃ temperature-programmed desorption (NH₃-TPD) were also performed on a Micromeritics Autochem II 2920 chemisorber equipped with a TCD and MS. For CO₂-TPD experiments, the sample (100 mg) was pretreated in a quartz U-tube reactor with a He gas stream (30 mL min^{-1}) at 473 K for 1 h. When the temperature decreased to 323 K, the CO₂ saturation uptake of the pretreated catalysts was achieved by passing pure CO₂ for 0.5 h at 30 mL min^{-1} . After the CO₂ adsorption, the catalyst was purged with Ar flow (40 mL min^{-1}) for 30 min at 323 K. Then the temperature was linearly increased from 323 K to 1073 K (10 K min^{-1}) in the Ar flow while the desorbed species were monitored using the TCD and MS in real time. NH₃-TPD experiments were carried out by a similar procedure to that of CO₂-TPD experiments except for using 10% NH₃/Ar gas mixture as the adsorbate.

Catalysts evaluation

The catalytic tests were performed in a fixed-bed reactor. Typically, 1 g of the catalyst was placed in a tubular reactor. The reaction was carried out under the following conditions: 523 K, 3 MPa, LHSV = $2 \text{ mL g}_{\text{cat}}^{-1} \text{ h}^{-1}$ and N₂/ethanol = 250:1 (v/v). The liquid products were analyzed by a gas chromatograph (GC) with a flame ionization detector (FID) and an HP-5 column (30 m, 0.25 mm inner diameter). 1-pentanol was used as the internal standard for the quantification of liquid products. The outlet gases from the condenser were analyzed using the GC with a TCD and an HP-PLOT/Q column (30 m, 0.32 mm inner diameter).

The ethanol conversion, selectivity and yield of products were calculated as follows:

$$\text{conversion (\%)} = \frac{C \text{ mol of products}}{C \text{ mol of (products + unreacted ethanol)}} \times 100\%$$

$$\text{selectivity (\%)} = \frac{C \text{ mol of specific product}}{C \text{ mol of products}} \times 100\%$$

$$\text{Product yield (\%)} = \text{Ethanol conversion} \times \text{Product selectivity}$$

where C mol is the mole number of carbon in the products and unreacted ethanol.

Results and discussion

XRD analysis of the NiO–Cu–La₂O₃/Al₂O₃ catalysts

The XRD patterns of this series of catalysts after the reaction are displayed in Fig. 1. The diffraction peaks at $2\theta = 43.3^\circ$, 50.4° and 74.1° are characteristic of the Cu crystal phase, while the others are attributed to the γ -alumina support. It can be found that the nickel-modified Cu–La₂O₃/Al₂O₃ catalysts except for the 2.55%NiO–La₂O₃/Al₂O₃ catalyst exhibited relatively weak metal Cu diffraction peaks in comparison with the Cu–La₂O₃/Al₂O₃ catalyst, suggesting that the addition of an appropriate amount of nickel improved the dispersion of copper; moreover, the intensity of metal Cu diffraction peaks reached the lowest at 0.65% of the NiO content, indicating the highest Cu dispersion for 0.65%NiO–Cu–La₂O₃/Al₂O₃. As confirmed by the results of the FTIR characterization of the Al₂O₃ support and the CuO/Al₂O₃, NiO/Al₂O₃ and CuO–NiO/Al₂O₃ catalysts with different CuO and/or NiO loadings (Fig. S1–S3 and Tables S1–S3[†]), both

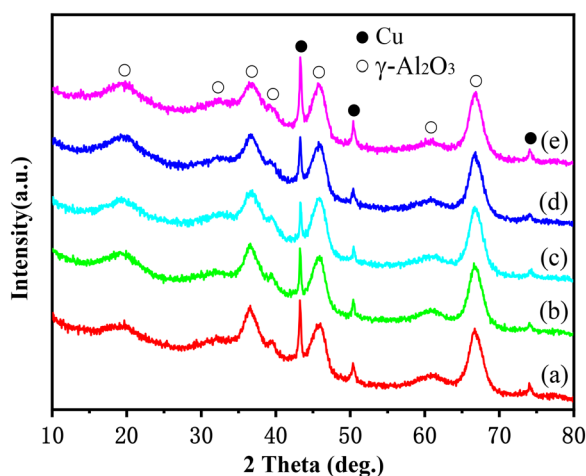


Fig. 1 XRD patterns of catalysts after reaction: (a) Cu–La₂O₃/Al₂O₃, (b) 0.32%NiO–Cu–La₂O₃/Al₂O₃, (c) 0.65%NiO–Cu–La₂O₃/Al₂O₃, (d) 1.27%NiO–Cu–La₂O₃/Al₂O₃, and (e) 2.55%NiO–Cu–La₂O₃/Al₂O₃.

CuO and NiO preferentially bonded with the coordinatively unsaturated Al³⁺,^{23,24} and then –OH on the surface of the Al₂O₃ support.²⁵ Thus, the initial enhancement in Cu dispersion with the increase of NiO loading might be attributed to the separation effect of the NiO promoter on CuO species on the surface of catalysts, while the subsequent decrease in Cu dispersion with the excessive loading of NiO should be ascribed to the overweight of competitive deposition of CuO and NiO over the separation effect of the NiO promoter on CuO species. Additionally, the characteristic diffraction peaks of metal Ni, NiO, La₂O₃ or other crystal phases of nickel and lanthanum species were not observed in the XRD patterns, indicating that the nickel and lanthanum species over these catalysts were highly dispersed or existed in amorphous forms.

TEM analysis of the NiO–Cu–La₂O₃/Al₂O₃ catalysts

As shown in Fig. 2A–D, there are no distinguishable metal Ni, NiO or La₂O₃ or other nanoparticles of nickel and lanthanum species except Cu nanoparticles found in typical HRTEM images of Cu–La₂O₃/Al₂O₃ and 0.65%NiO–Cu–La₂O₃/Al₂O₃ catalysts, suggesting that the nickel and lanthanum species were highly dispersed or existed in amorphous forms, which is consistent with the XRD results. The HRTEM and the

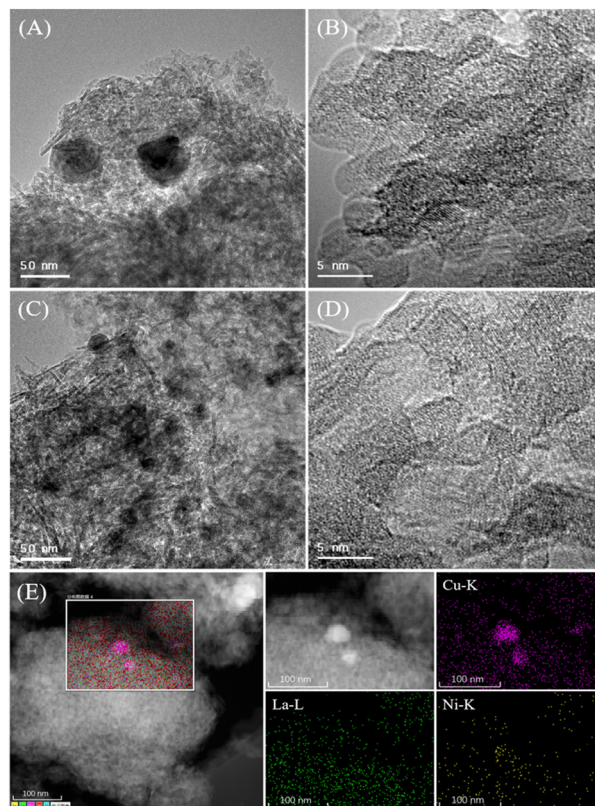


Fig. 2 (A) and (B) HRTEM images of the Cu–La₂O₃/Al₂O₃ catalyst after reaction, (C) and (D) HRTEM images, and (E) HRTEM and the corresponding EDX mapping images of the 0.65%NiO–Cu–La₂O₃/Al₂O₃ catalyst after reaction.

corresponding EDX mapping images (Fig. 2E) further suggest that a small amount of copper existed in the form of Cu nanoparticles, while most of the Cu species were ultra-highly dispersed. In particular, the EDX mapping photographs show that nickel and lanthanum species were not only highly dispersed but also contacted intimately with Cu species or partially covered Cu nanoparticles which may play an important role in preventing the sintering of the ultra-highly dispersed Cu species or Cu nanoparticles. Additionally, the average size of the Cu nanoparticles over the 0.65%NiO-Cu-La₂O₃/Al₂O₃ catalyst was smaller than that over the Cu-La₂O₃/Al₂O₃ catalyst (25 vs. 42 nm), further confirming that the addition of nickel oxide improved the dispersion of Cu species effectively.

H₂-TPR analysis of the NiO-Cu-La₂O₃/Al₂O₃ catalysts

The H₂-TPR profiles of this series of catalysts are shown in Fig. 3. It can be seen that each sample showed two H₂ consumption peaks. The peaks at low temperature were attributed to the reduction of CuO species on the surface of these catalysts,^{26–28} while the H₂ consumption peaks in the range of 800 K to 1000 K probably corresponded to the reduction of Cu-Al mixed oxides such as CuAl₂O₄ or CuAlO₂.^{29,30} Area of the peak at low temperature first increased with the NiO content from 0 to 0.65% and then decreased with the further increase of the NiO content to 2.55%. The maximum area of the H₂ consumption peak was observed for the 0.65%NiO-Cu-La₂O₃/Al₂O₃ catalyst, indicating the maximum amount of reduced CuO species over the surface of this catalyst. Moreover, the reduction peaks at both low and high temperatures shifted to a higher temperature with the increase of the NiO content, implying the existence of a strong interaction between Cu and Ni species, which will improve the stability of Cu species. Additionally, it can be found from the H₂-TPR profile of the

2.55%NiO/Al₂O₃ catalyst (Fig. S4†) that the H₂ consumption peak of NiO species on the surface of NiO/Al₂O₃ catalyst ranged from 750 K to 1000 K, with the maximum at 851 K, while the NiO reduction peak shifted to high temperature with the addition of lanthanum and almost overlapped with the H₂ consumption peak of Cu-Al spinel on the Cu-La₂O₃/Al₂O₃ and NiO-Cu-La₂O₃/Al₂O₃ catalysts. Furthermore, the intensity of the high temperature peak in the TPR spectra of the NiO-Cu-La₂O₃/Al₂O₃ catalysts slightly increased with the increase of NiO loading (Fig. 3). These results indicated that the reduction peaks of NiO or Ni-Al spinel species on the NiO-Cu-La₂O₃/Al₂O₃ catalysts were probably included in the high temperature peaks and overlapped with the H₂ consumption peaks of the Cu-Al spinels.

XPS analysis of the NiO-Cu-La₂O₃/Al₂O₃ catalysts

In order to further identify the chemical states of Ni and Cu species, the Cu-La₂O₃/Al₂O₃ and 0.65%NiO-Cu-La₂O₃/Al₂O₃ catalysts after reaction was characterized by XPS. As shown in Fig. 4A, the peaks at 855.5 and 873.8 eV along with the satellite peaks at 862.6 and 879.1 eV are respectively characteristic of Ni²⁺ 2p_{3/2} and 2p_{1/2} in the form of NiO, while there were no peaks characteristic of metallic Ni (at ca. 853.0 eV) found in the Ni 2p XPS spectrum.^{31–33} These results indicate that the nickel species on the surface of the catalyst mainly existed in the form of NiO. Additionally, the peaks at 852.0 and 856.7 eV were ascribed to La³⁺ 3d_{3/2}, while the ones at 838.5 and 835.6 eV were ascribed to La³⁺ 3d_{5/2} (Fig. 4A and S5 and Table S4†).³⁴ Furthermore, the binding energies of La 3d_{5/2} (835.6 eV) were substantially higher than the values measured for bulk La₂O₃ (833.2 eV) standards, indicating a high dispersion of La₂O₃ on the surface of the catalysts,³⁵ which was consistent with the former XRD and HRTEM results. As shown in Fig. 4B, the Cu-La₂O₃/Al₂O₃ and 0.65%NiO-Cu-La₂O₃/Al₂O₃ catalysts exhibited a similar Cu 2p XPS spectrum in which two peaks at ca. 932.1 and 952.3 eV corresponded to Cu⁺ or Cu⁰,^{36–38} while the peaks located at ca. 933.4 and 953.4 eV with a satellite peak at 940–945 eV confirmed the existence of Cu²⁺.^{39,40} The Cu LMM AES spectrum of the catalysts (Fig. 4C) further confirmed that the low-valence Cu species comprised two kinds of Cu species, i.e., Cu⁺ and Cu⁰. The ratios of Cu²⁺, Cu⁺ and Cu⁰ were calculated to be 1.2:1.2:1 and 2.6:5.6:1 for the Cu-La₂O₃/Al₂O₃ and 0.65%NiO-Cu-La₂O₃/Al₂O₃ catalysts, respectively, while the ratio of the total amount of Cu species detected in the two catalysts was 0.96:1, indicating that the addition of NiO suppressed the progressive reduction of Cu²⁺ → Cu⁺ → Cu⁰ and increased significantly the relative amount of the Cu⁺ species. This phenomenon might be explained as follows: 1) the existence of a strong interaction between Cu and Ni species as confirmed by the XRD results; 2) the creation of more vacancies by adding NiO on the surface of the catalyst, which was beneficial to the stabilization of Cu⁺ active species.⁴¹ Ramasamy *et al.* concluded that the presence

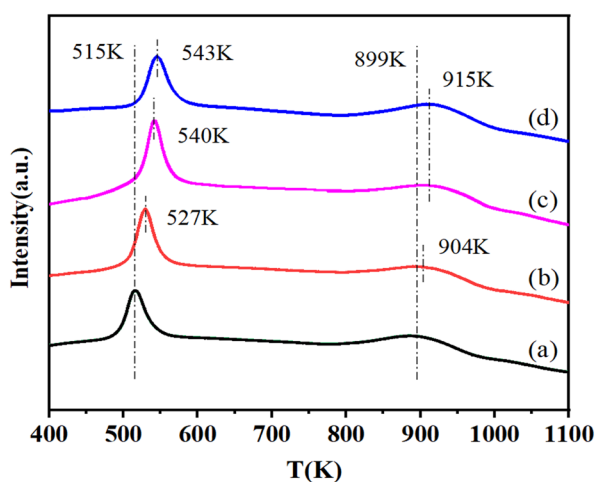


Fig. 3 H₂-TPR profiles of calcined catalysts: (a) Cu-La₂O₃/Al₂O₃, (b) 0.32%NiO-Cu-La₂O₃/Al₂O₃, (c) 0.65%NiO-Cu-La₂O₃/Al₂O₃, and (d) 2.55%NiO-Cu-La₂O₃/Al₂O₃.

of the atomically dispersed Cu^{1+} species over the low copper loaded MgAl mixed oxide catalysts for conversion of ethanol to higher alcohols is a key factor in achieving a high dehydrogenation rate while minimizing side product formation.⁴² In our recent study, octahedrally coordinated Cu^{2+} was demonstrated to be highly active for dehydrogenation/hydrogenation during ethanol upgrading.¹⁸

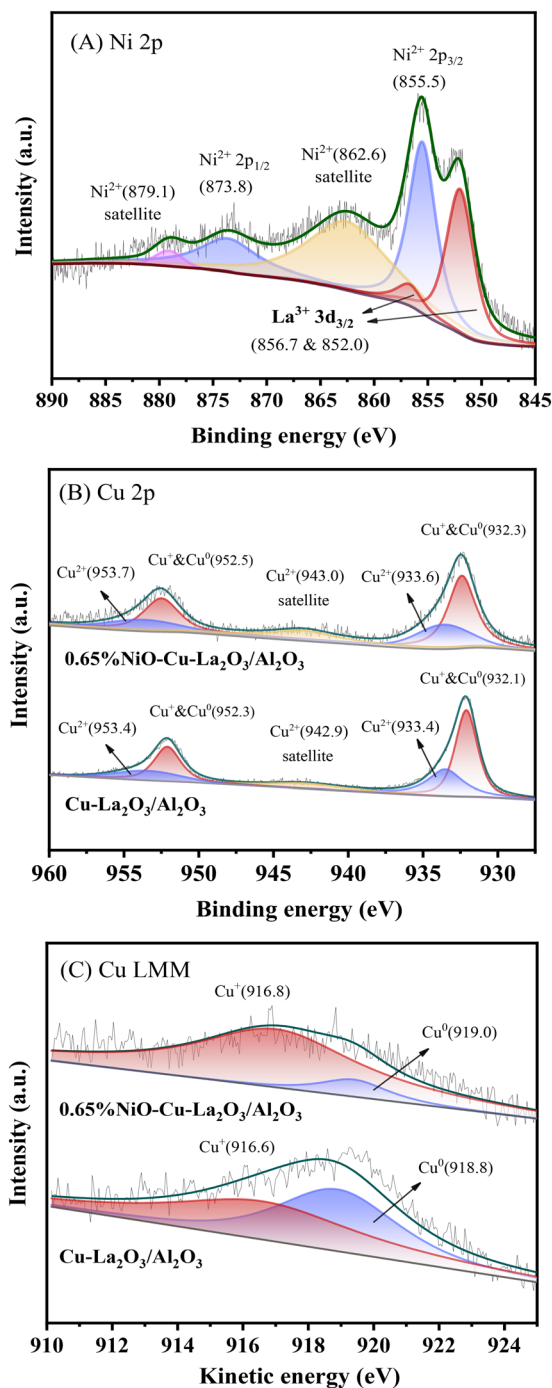


Fig. 4 (A) Ni 2p XPS spectrum of the 0.65%NiO-Cu-La₂O₃/Al₂O₃ catalyst after the reaction. (B) Cu 2p XPS spectra and (C) Cu LMM AES spectra of the Cu-La₂O₃/Al₂O₃ and 0.65%NiO-Cu-La₂O₃/Al₂O₃ catalysts after the reaction.

Furthermore, Cu^0 has been generally considered the active center of ethanol dehydrogenation and crotonaldehyde hydrogenation.²² Thus, all of Cu^{2+} , Cu^+ and Cu^0 should be active sites for the dehydrogenation and hydrogenation steps during the Guerbet coupling of ethanol over the NiO-Cu-La₂O₃/Al₂O₃ catalyst. Moreover, the increase of the total amount of Cu^{2+} , Cu^+ and Cu^0 on the surface of the 0.65%NiO-Cu-La₂O₃/Al₂O₃ catalyst compared with the Cu-La₂O₃/Al₂O₃ catalyst implied that the addition of an appropriate amount of NiO increased the amount of Cu active sites, which was consistent with the former XRD and TEM results.

H₂-TPD analysis of the NiO-Cu-La₂O₃/Al₂O₃ catalysts

The H₂-TPD profiles of this series of catalysts are presented in Fig. 5. Two H₂ desorption peaks at *ca.* 364 K and 728 K were observed for all samples. The peak at low temperature was attributed to desorption of H₂ adsorbed on the Cu active sites, while the peak at high temperature could be attributed to the desorption of hydrogen spilled over lanthanum oxide or an Al₂O₃ support.^{43,44} The amount of H₂ desorbed from Cu active sites increased as follows: Cu-La₂O₃/Al₂O₃ (0.058 mmol g⁻¹) < 0.32%NiO-Cu-La₂O₃/Al₂O₃ (0.104 mmol g⁻¹) < 2.55%NiO-Cu-La₂O₃/Al₂O₃ (0.233 mmol g⁻¹) < 0.65%NiO-Cu-La₂O₃/Al₂O₃ (0.330 mmol g⁻¹), implying the trend of the capability for hydrogen adsorption and activation. As confirmed by XRD, H₂-TPR and XPS results, 0.65%NiO-Cu-La₂O₃/Al₂O₃ catalyst had the highest Cu dispersion and the largest amount of Cu active species, which were responsible for the highest capability for hydrogen activation.^{45,46} Though the amounts of the spilled hydrogen on the NiO-containing catalysts were less than those on the Cu-La₂O₃/Al₂O₃ catalyst, the spilled hydrogen had lower desorption temperature, probably indicating that the addition of NiO increased the activity of the spilled hydrogen.

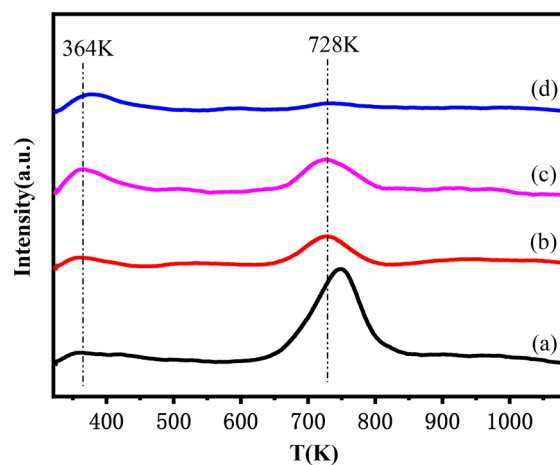


Fig. 5 H₂-TPD profiles of the reduced catalysts: (a) Cu-La₂O₃/Al₂O₃, (b) 0.32%NiO-Cu-La₂O₃/Al₂O₃, (c) 0.65%NiO-Cu-La₂O₃/Al₂O₃, and (d) 2.55%NiO-Cu-La₂O₃/Al₂O₃.

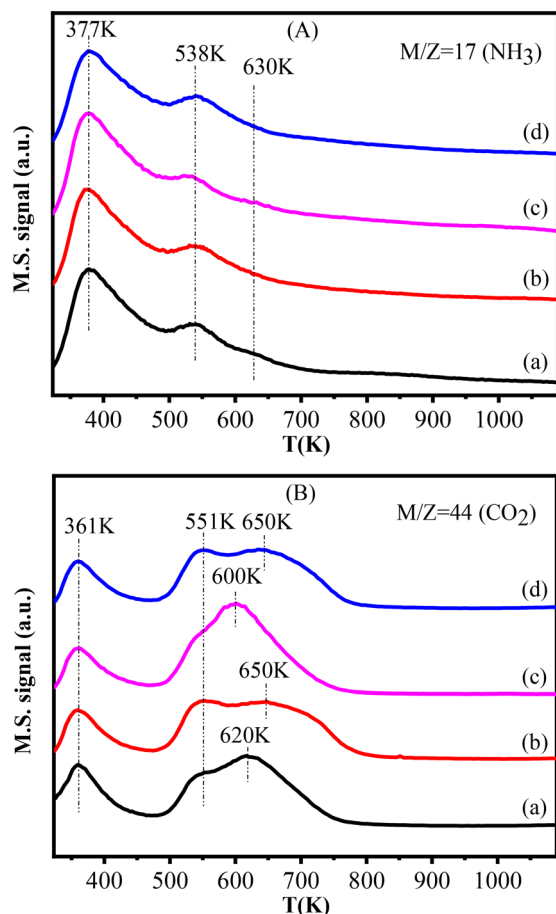


Fig. 6 NH_3 -TPD (A) and CO_2 -TPD (B) profiles of the catalysts: (a) $\text{Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$, (b) $0.32\%\text{NiO-Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$, (c) $0.65\%\text{NiO-Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$, and (d) $2.55\%\text{NiO-Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$.

NH_3 -TPD and CO_2 -TPD analysis of the $\text{NiO-Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$ catalysts

The NH_3 -TPD and CO_2 -TPD profiles of this series of catalysts are presented in Fig. 6. The $\text{Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$ catalyst exhibited three NH_3 desorption peaks centered at 377 K, 538 K and 630 K (Fig. 6A), which were assigned to the weak, medium and strong acidic sites, respectively.⁴⁷ These acidic sites were related to Al^{3+} with different coordination structural patterns and aluminum hydroxyl groups on the surface of the Al_2O_3 support.^{48–50} According to Fig. 6A and Table S5,[†] the strong acidic sites disappeared, while the total amount of weak and medium acidic sites slightly decreased with the addition of NiO, which is understandable given the weak alkalinity of nickel oxide.

As shown in Fig. 6B, there were three CO_2 desorption peaks at 361 K, ~550 K and 600–650 K observed for each catalyst. The first CO_2 desorption peak corresponding to weak basic sites was associated to the surface basic hydroxyl groups ($-\text{OH}$), while the subsequent two peaks corresponding respectively to medium strength and strong basic sites might be related to metal–oxygen pairs (Al-O and La-O) and low coordination surface O^{2-} .^{51–55} It can be seen from Fig. 6B and

Table S5[†] that the amount of the weak basic sites almost did not change, while that of the medium and strong basic sites slightly increased after introducing nickel oxide. In particular, for the $0.65\%\text{NiO-Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$ catalyst (Fig. 6B(c)), the number of strong bases was the largest, and the distribution was most concentrated, probably implying the highest and most uniform dispersion of La species.^{54,55} Thus, the addition of an appropriate amount of NiO optimized the acid–base distribution and promoted the balance of acid–base sites, which is essential for aldol condensation of acetaldehyde, thus improving the efficiency of catalysts for ethanol upgrading to higher alcohols.

Catalytic performance of the $\text{NiO-Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$ catalyst

The catalytic performances of this series of catalysts for upgrading of ethanol to higher alcohols are presented in Table 1. As shown in Table 1, $\text{Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$ exhibited 50.9% ethanol conversion, 67.9% total selectivity to higher alcohols (S_T), and 34.6% total yield thereof (Y_T). The ethanol conversion and Y_T first increased and then decreased with the increase of NiO. The optimal $0.65\text{wt}\%\text{NiO-Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$ catalyst exhibited an ethanol conversion of 56.7%, a S_T of 76.1%, and Y_T of up to 43.1%. What's more remarkable was that the $0.65\%\text{NiO-Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$ catalyst exhibited excellent stability: ethanol conversion, S_T and Y_T remained almost the same during 200 h on-stream evaluation (Fig. S6[†]). Additionally, the liquid yield of the $\text{Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$ catalyst was 99.3%, while those of the $\text{Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$ catalysts modified with 0.32%, 0.65%, and 1.27% NiO were 98.3%, 98.0% and 97.8%, respectively. The cracking rates of liquid products calibrated by decreasing the liquid loss during the collection of samples were below 1.5%, indicating that the highly dispersed NiO species on the $\text{NiO-Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$ catalysts with NiO loadings of 0.32%, 0.65%, and 1.27% had extremely low activity of cracking the C–C bond. However, the liquid yield of the $2.55\%\text{NiO-Cu-La}_2\text{O}_3/\text{Al}_2\text{O}_3$ catalyst was 95.8%, corresponding to the cracking rate of 3.5%, which might be caused by the reduction of a small amount of nickel oxide to metallic nickel during ethanol upgrading.^{19–21} Since NiO itself showed no catalytic activity for catalytic upgrading of ethanol to higher alcohols (Table S6[†]), the introduction of an appropriate amount of NiO actually increased the amount of Cu active sites and therefore enhanced the ethanol dehydrogenation activity of the Cu-based catalyst, which can be reflected by the increase of ethanol conversion and Y_T with the addition of NiO. The increase of the amount of Cu active sites also enhanced the hydrogen adsorption and activation ability of the Cu-based catalyst, as directly confirmed by the H_2 -TPD results. Furthermore, the nickel species in the form of NiO might be favorable for the adsorption and activation of esters and aldehydes by interacting with their oxygen atoms on carbonyl groups^{56,57} and promote their reduction to alcohols by the hydrogen from adjacent Cu active sites. Besides ethanol dehydrogenation and hydrogenation of crotonaldehyde,

Table 1 Catalytic performance of the NiO–Cu–La₂O₃/Al₂O₃ catalysts for ethanol upgrading to higher alcohols^a

Catalyst	Conv. (%)	Sel. (%)						<i>S_T</i> ^d	<i>Y_T</i> ^e
		Acetaldehyde	Ethyl acetate	Butyraldehyde	Butanol	C6–C8 alcohols ^b	Others ^c		
Cu–La ₂ O ₃ /Al ₂ O ₃	50.9	1.7	6.5	4.2	48.6	19.3	19.7	67.9	34.6
0.32%NiO–Cu–La ₂ O ₃ /Al ₂ O ₃	55.3	1.8	3.1	3.8	48.5	26.1	16.7	74.6	41.3
0.65%NiO–Cu–La ₂ O ₃ /Al ₂ O ₃	56.7	1.9	3.1	3.5	49.7	26.4	15.4	76.1	43.1
1.27%NiO–Cu–La ₂ O ₃ /Al ₂ O ₃	55.9	1.5	2.7	3.3	49.8	27.1	15.6	76.9	43.0
2.55%NiO–Cu–La ₂ O ₃ /Al ₂ O ₃	51.3	1.6	2.1	3.1	50.5	28.9	13.8	79.4	40.7

^a Conversion and selectivity are obtained at steady-state; reaction conditions: catalyst, 1.0 g; 523 K, 3 MPa (N₂), LHSV = 2 ml g_{cat}^{−1} h^{−1}, N₂/ethanol (v/v) = 250:1. ^b C6–C8 alcohols include 2-ethylbutanol, 1-hexanol, 2-ethylhexanol and 1-octanol. ^c Other products include diethyl ether, 1,1-diethoxyethane, butyl acetate, etc. ^d *S_T* represents the total selectivity to higher alcohols. ^e *Y_T* represents the total yield to higher alcohols.

acetaldehyde condensation is the other key elementary step in the ethanol to *n*-butanol process, which is highly dependent on the acidic and basic properties of catalysts. Moreover, besides solo acid or base sites, acid–base pairs demonstrated elevated activity in condensation reactions.^{18,58}

As confirmed by the results of NH₃-TPD and CO₂-TPD, the addition of NiO reduced slightly the number of acidic sites on the surface of the Cu-based catalyst while increased the amount of basic sites, especially strong basic sites, forming abundant acid–base pairs, which promoted the acetaldehyde condensation reactions and then the ethanol upgrading to higher alcohols.

In summary, the catalytic performance of the NiO–Cu–La₂O₃/Al₂O₃ catalysts were obviously improved by the addition of nickel oxide, and the 0.65%NiO–Cu–La₂O₃/Al₂O₃ catalyst exhibited excellent catalytic activity and the highest total yield of higher alcohols in continuous catalytic upgrading of ethanol. The addition of an appropriate NiO amount effectively not only improved the dehydrogenation/hydrogenation ability of the catalyst by improving the dispersion of Cu species, but also optimized the acid–base properties of the catalysts, therefore promoting the aldol condensation of acetaldehyde.

Conclusions

A series of NiO–Cu–La₂O₃/Al₂O₃ catalysts with varying NiO contents (0–2.55 wt%) were prepared using a simple wetness impregnation method and applied for the catalytic upgrading of ethanol to higher alcohols. The catalytic performance of the Cu–La₂O₃/Al₂O₃ catalyst was obviously influenced by addition of NiO, and the optimal 0.65%NiO–Cu–La₂O₃/Al₂O₃ catalyst exhibited 56.7% ethanol conversion, 76.1% total selectivity to higher alcohols and up to 43.1% total yield thereof. The characterization results proved that addition of an appropriate amount of NiO not only increased the amount of surface Cu active species, thus improving the dehydrogenation/hydrogenation capability of the catalyst, but also adjusted the acid–base properties of the catalysts, thus facilitating aldol condensation of acetaldehyde. This study demonstrated that the introduction of metal oxide promoters with specific adjustment functions can be a feasible

approach to achieve more efficient Cu-based catalysts for catalytic upgrading of ethanol to higher alcohols.

Author contributions

Jinfeng He: investigation, resources, analysis, data curation, and writing – original draft; Xiuzhen Li: investigation and data curation; Jianyao Kou: investigation, analysis, and resources; Tingjie Tao: investigation and resources; Xinyue Shen: investigation and data curation; Dahao Jiang: writing – review & editing, project administration, supervision, and funding acquisition; Lili Lin: resources and analysis; Xiaonian Li: supervision and project administration.

Conflicts of interest

There are no conflicts to declare.

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